

DINOv2 ablation results — Imagenette & HCP

Loss curves and linear-probe scores for every run. Part 1: Imagenette (control on natural images). Part 2: HCP brain MRI.

Metric. All scores are a linear probe on frozen features. The loss curve is shown for reference only — a flat or rising loss can still give the best probe, and a loss diving to 0 indicates collapse.

Part 1 Imagenette — control on natural images

9,469 images, 10 classes. Probe = linear classification of the 10 classes (validation accuracy, chance = 10%).

TEST 1 · DEFAULT CONFIGURATION

DINOv2 out of the box

9,469 images

65,536 prototypes

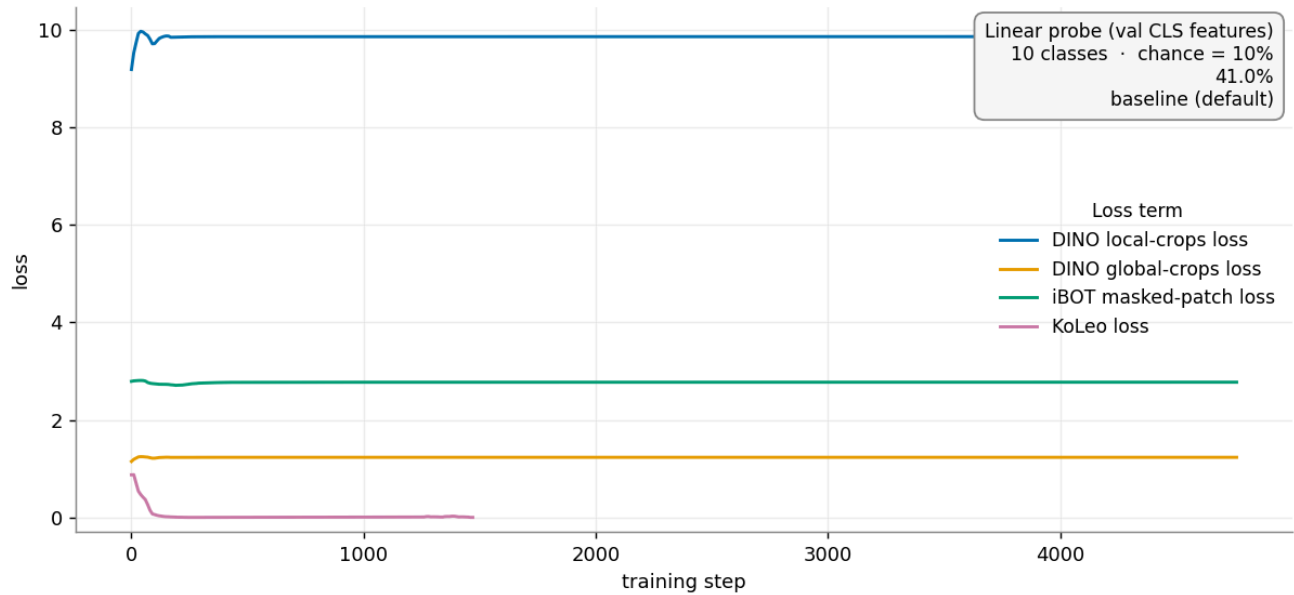
teacher_temp 0.04

ko leo on

Original ImageNet-scale hyperparameters, unchanged.

RESULT Loss flat throughout; probe 41% (chance 10%).

Default config (full Imagenette)
9469 images · teacher_temp 0.04 · 65536 prototypes · koleo weight = 0.1



TEST 2 · PROTOTYPES + TEMPERATURE

Fewer prototypes, sharpened temperature

4,096 prototypes

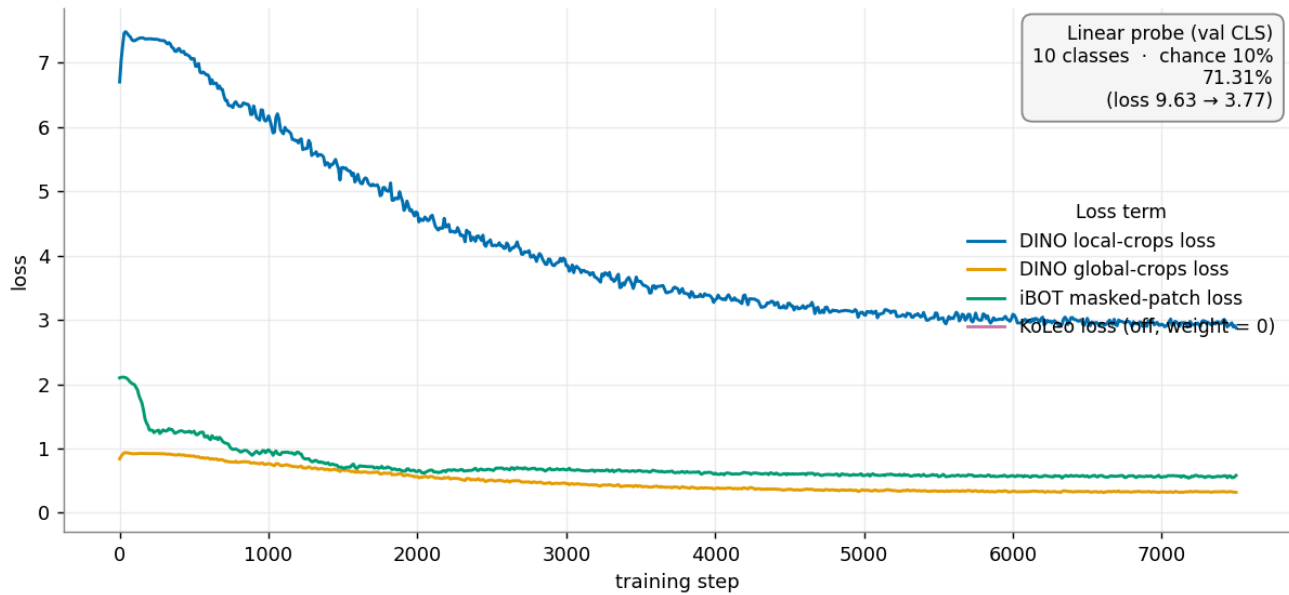
teacher_temp 0.04 → 0.02

koleo = 0

Prototypes cut to 4,096, teacher temperature sharpened toward the end (0.04 → 0.02), koleo disabled.

RESULT Loss drops 9.6 → 3.8; probe 71.3% (+30 vs default).

Imagenette — teacher_temp 0.04 → 0.02 (finishes SHARP)
9469 images · 4096 prototypes · koleo=0 · standard crops



Part 2 HCP — brain MRI

Each fMRI scan → 2D axial brain slices. Probe = sex prediction (**chance = 50%**), 5-fold CV by subject. Two changes are tested alone and combined — **A**: full-view crops (local = global); **B**: 8 timepoints per scan (multi-frame). Everything else is fixed (4096 prototypes, koleo = 0, temp 0.01 → 0.04).

TEST 1 · BASELINE

One image per scan, standard crops

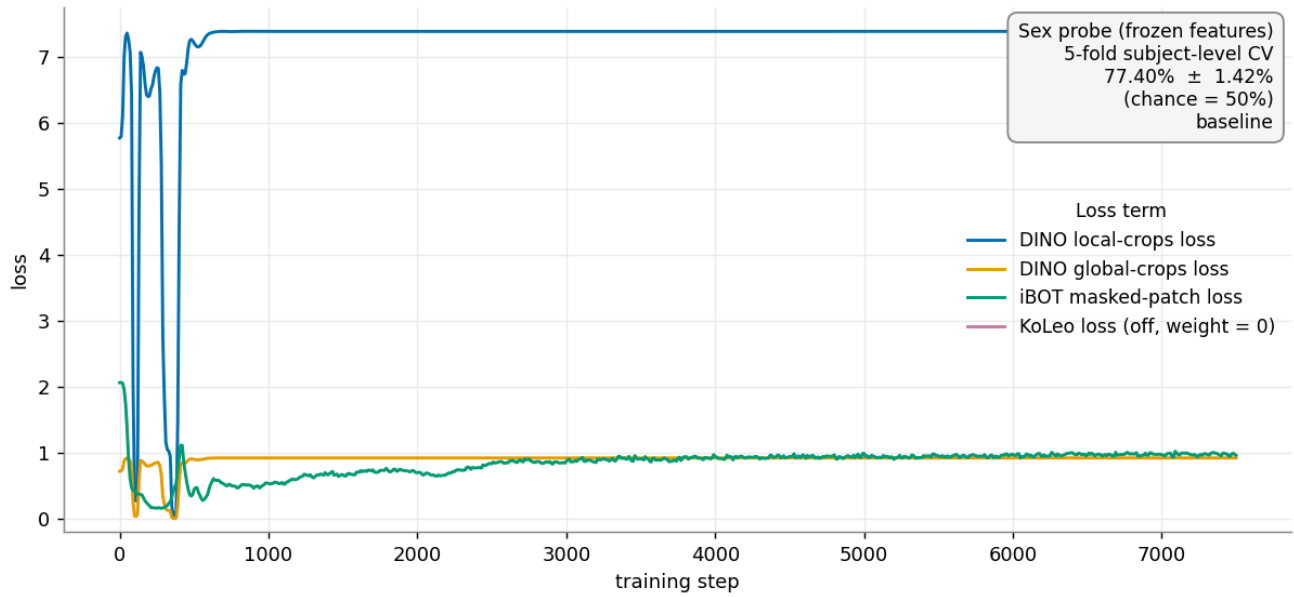
1 image / scan

standard crops 96px

One temporal-mean image per scan, standard DINO crops. The reference.

RESULT Probe 77.4% ± 1.4.

Baseline
1 temporal-mean image per scan · standard small local crops (96px) · koleo = 0



TEST 2 · CHANGE A

Full-view local crops (local = global)

1 image / scan

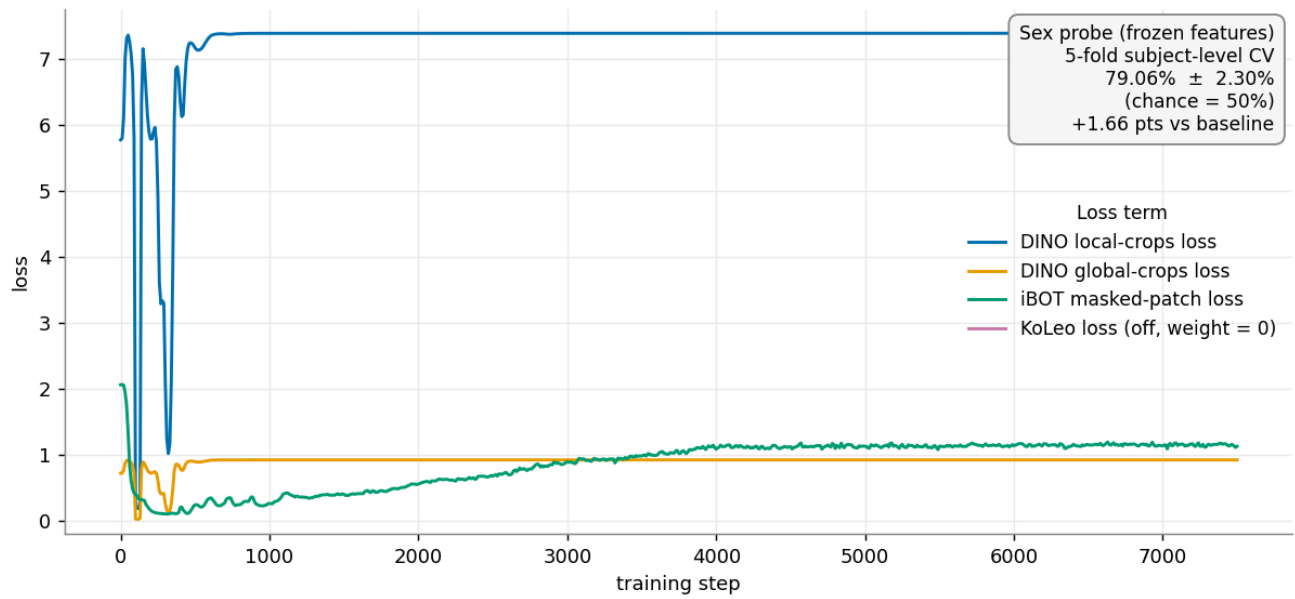
local = global

Local crops enlarged to full view instead of small 96px crops.

RESULT Probe 79.1% $\pm$ 2.3 · +1.7 vs baseline.

Ablation A — local crops = global

1 temporal-mean image per scan · local crops enlarged to full-view 224px (same as global) · koleo = 0



TEST 3 · CHANGE B

Eight timepoints per scan (multi-frame)

multi-frame (8 / scan)

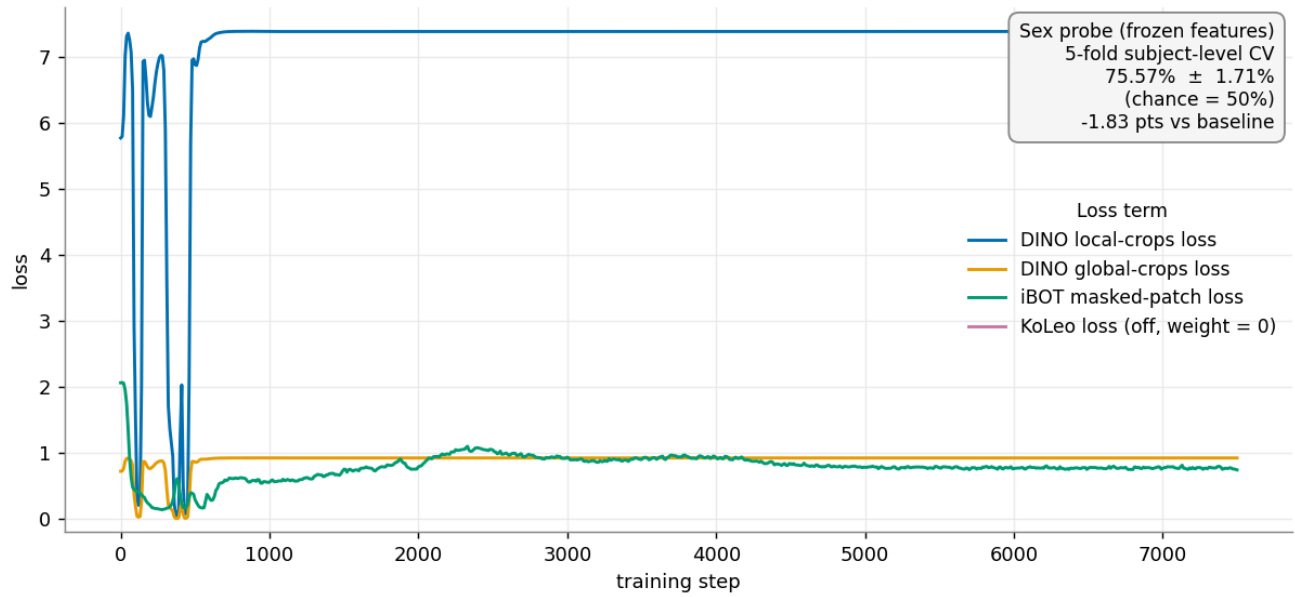
standard crops 96px

Eight slices at eight timepoints instead of one mean image.

RESULT Probe $75.6\% \pm 1.7$ · -1.8 vs baseline (hurts alone).

Ablation B — multi-frame

8 axial slices at 8 timepoints per scan · standard small local crops (96px) · koleo = 0



TEST 4 · A + B

Multi-frame + full-view crops

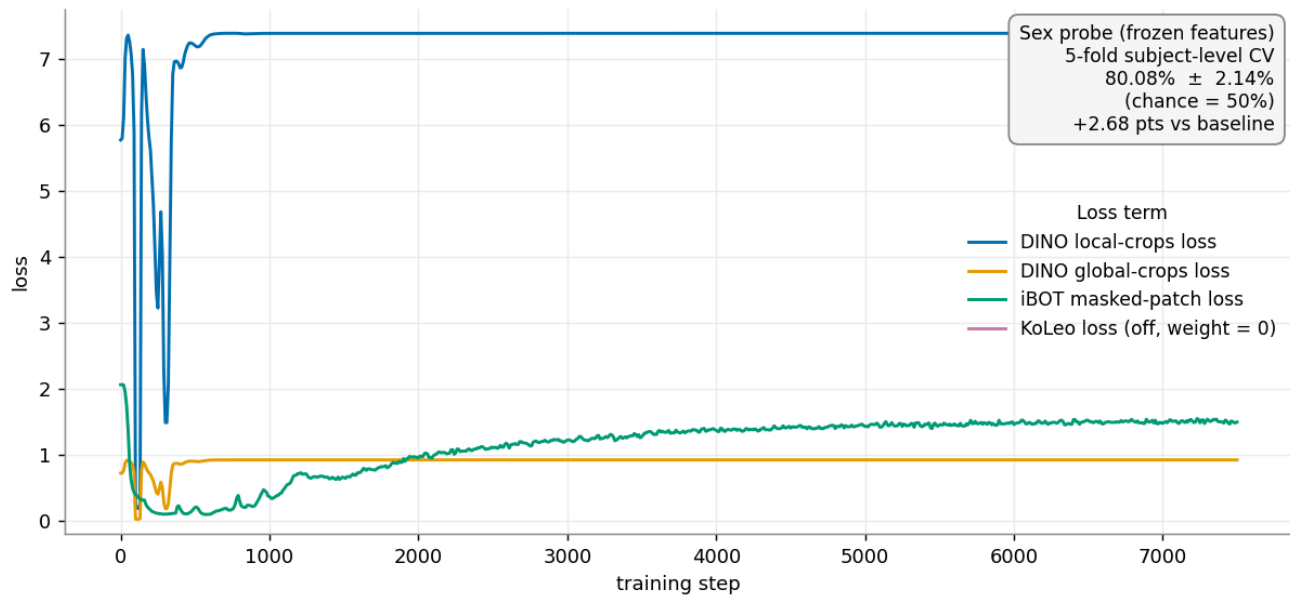
multi-frame (8 / scan)

local = global

Both changes together.

RESULT Probe $80.1\% \pm 2.1$ · best result.

Ablation A+B — multi-frame & local=global
8 timepoints per scan · local crops = global (224px) · 4096 prototypes · koleo = 0



Summary

Imagenette — the probe reacts strongly to hyperparameters: 41% $\rightarrow$ 71%.

HCP — full-view crops (A) help (+1.7); multi-frame (B) hurts on its own (−1.8); together they give the best result, 80.1%.

Next — the brain probe moves far less than Imagenette's. The likely lever is a real gap between the two views: different temporal windows per view ($\pm$ 3D augmentations).

Probe = linear classification on frozen features · HCP: sex, 5-fold CV by subject, chance 50% · Imagenette: 10 classes, chance 10% · All runs: 4096 prototypes, koleo = 0.